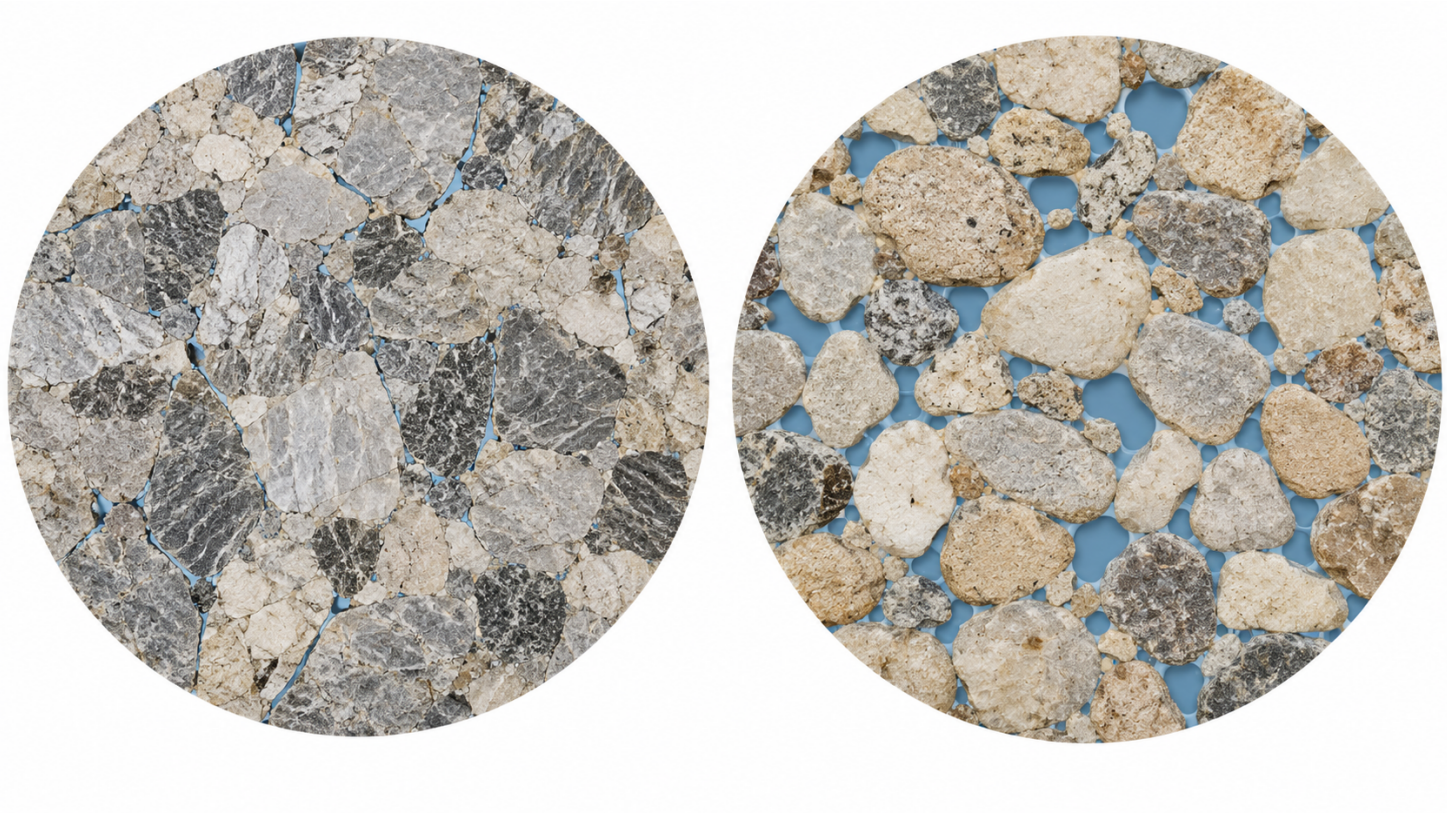
Void geometry matters as much as void fraction



COMPACT ROCK

crack-like voids · pressure-sensitive

POROUS ROCK

equant pores + throats · connectivity matters

GROUP CHALLENGE

Design the measurement before seeing the equations

Available

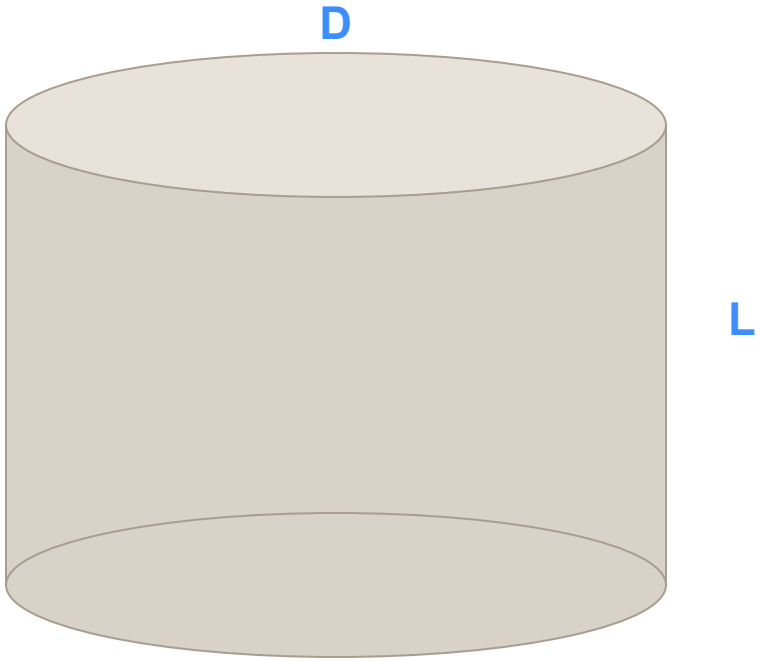
Scale	**caliper**	**water**	**dry + saturated core**

Propose a sequence that gives:

- bulk volume
- dry bulk density
- water-accessible porosity

Write the raw measurements first.
Do not begin with a formula.

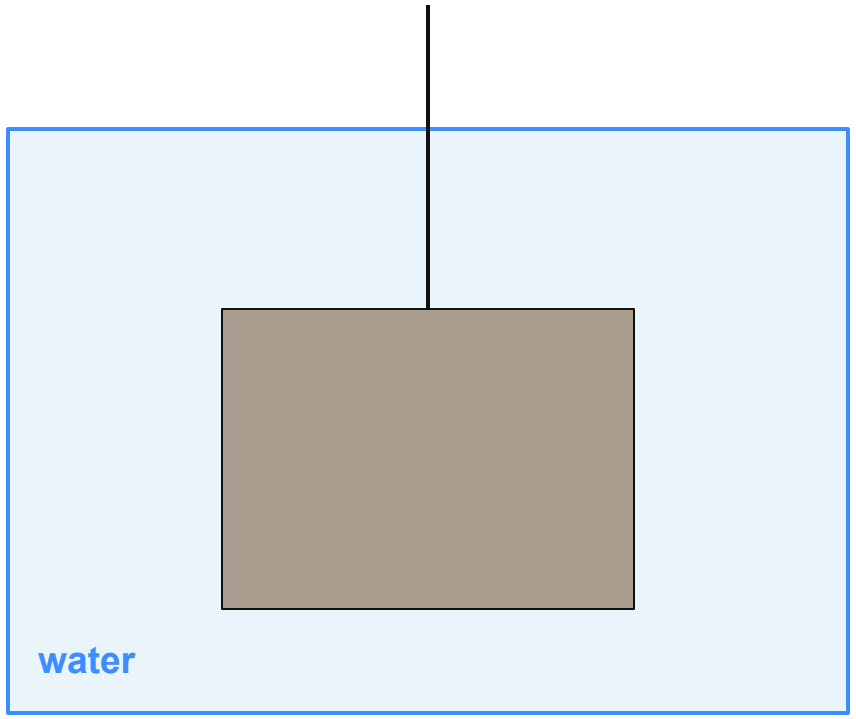
Geometry gives one estimate of bulk volume



$$V_{b,\text{geom}} = \frac{\pi D^2 L}{4}$$

Works best for regular specimens.
Surface roughness and diameter variation matter.

Buoyancy gives a second bulk-volume estimate



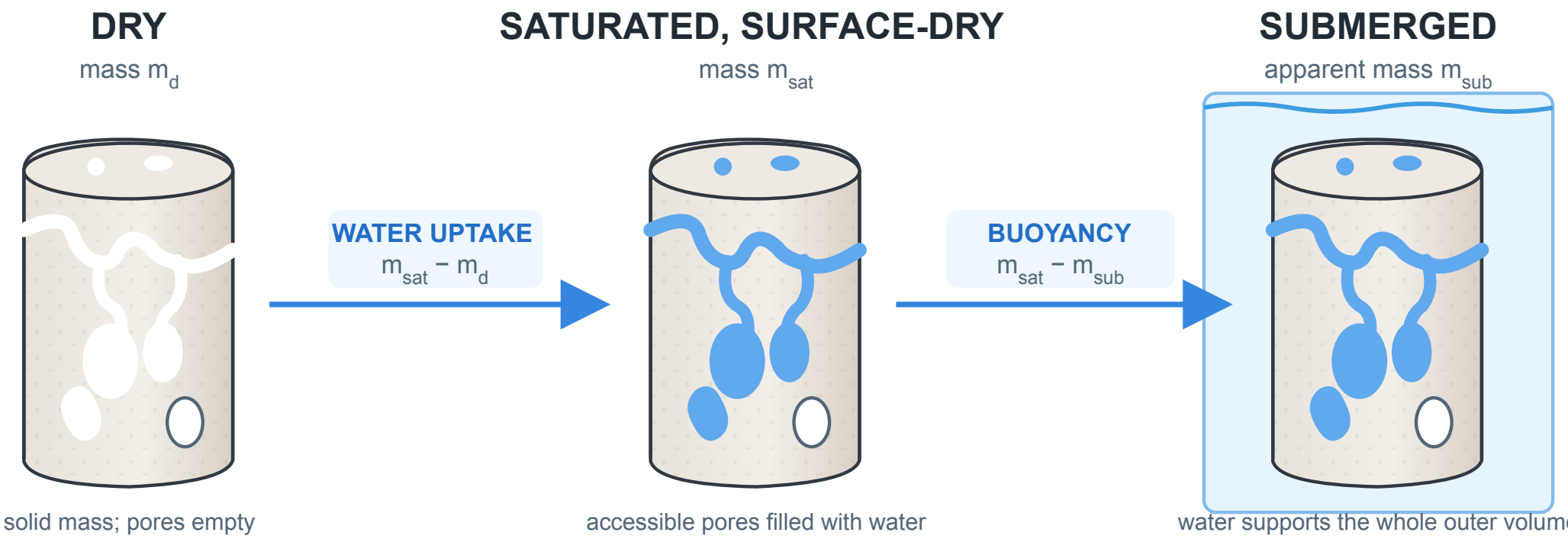
$$V_b = \frac{m_{\text{sat}} - m_{\text{sub}}}{\rho_w}$$

The apparent loss of mass equals the mass of displaced water.

When and why this method can be useful?

Remove bubbles · avoid wall contact · use ρw(T)

Three masses reveal water-accessible pore volume



$$V_{p,e} = \frac{m_{\text{sat}} - m_d}{\rho_w}$$

$$\rho_{b,d} = \frac{m_d \rho_w}{m_{\text{sat}} - m_{\text{sub}}}$$

$$\phi_e = \frac{m_{\text{sat}} - m_d}{m_{\text{sat}} - m_{\text{sub}}}$$

Three masses reveal water-accessible pore volume

m_d dry mass

solid mass

m_{sat} **saturated, surface-dry**

solid + accessible pore water

m_{sub} **apparent submerged mass**

buoyancy applied

$$V_{p,e} = \frac{m_{\text{sat}} - m_d}{\rho_w}$$

$$\rho_{b,d} = \frac{m_d \rho_w}{m_{\text{sat}} - m_{\text{sub}}}$$

$$\phi_e = \frac{m_{\text{sat}} - m_d}{m_{\text{sat}} - m_{\text{sub}}}$$

17 MINUTES

Mini-lab: measure → calculate → challenge

01	**Measure**	D, L, m_{sat}, m_{sub}
02	**Calculate**	V_β, ρ_β,d, φe
03	**Compare**	geometry vs buoyancy
04	**Challenge**	dominant uncertainty

GROUP ROLES

Experimentalist

Recorder

Skeptic

Record raw data before calculated values.

CLASS SYNTHESIS

A result is useful only when groups can compare it

Enter one row per group

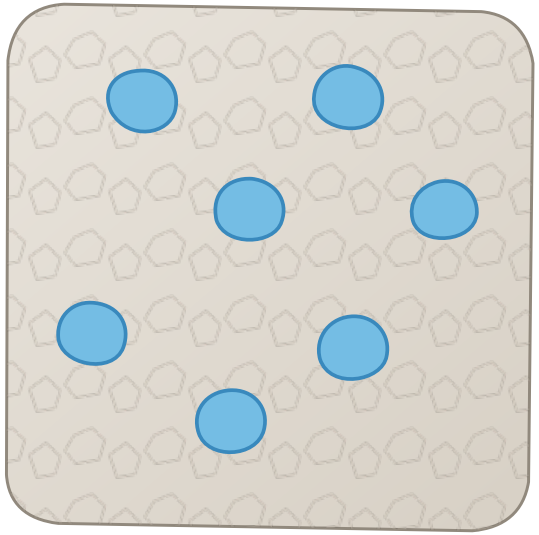
Sample	Vgeom	Vbuoy	$\rho_{\beta,d}$	φ_e	main uncertainty

Discuss: **Do the two bulk-volume methods agree within uncertainty?**

What explains disagreement between groups?

Equal porosity can still produce different behavior

Equant pores



Mostly isolated pore space

Thin, linked cracks

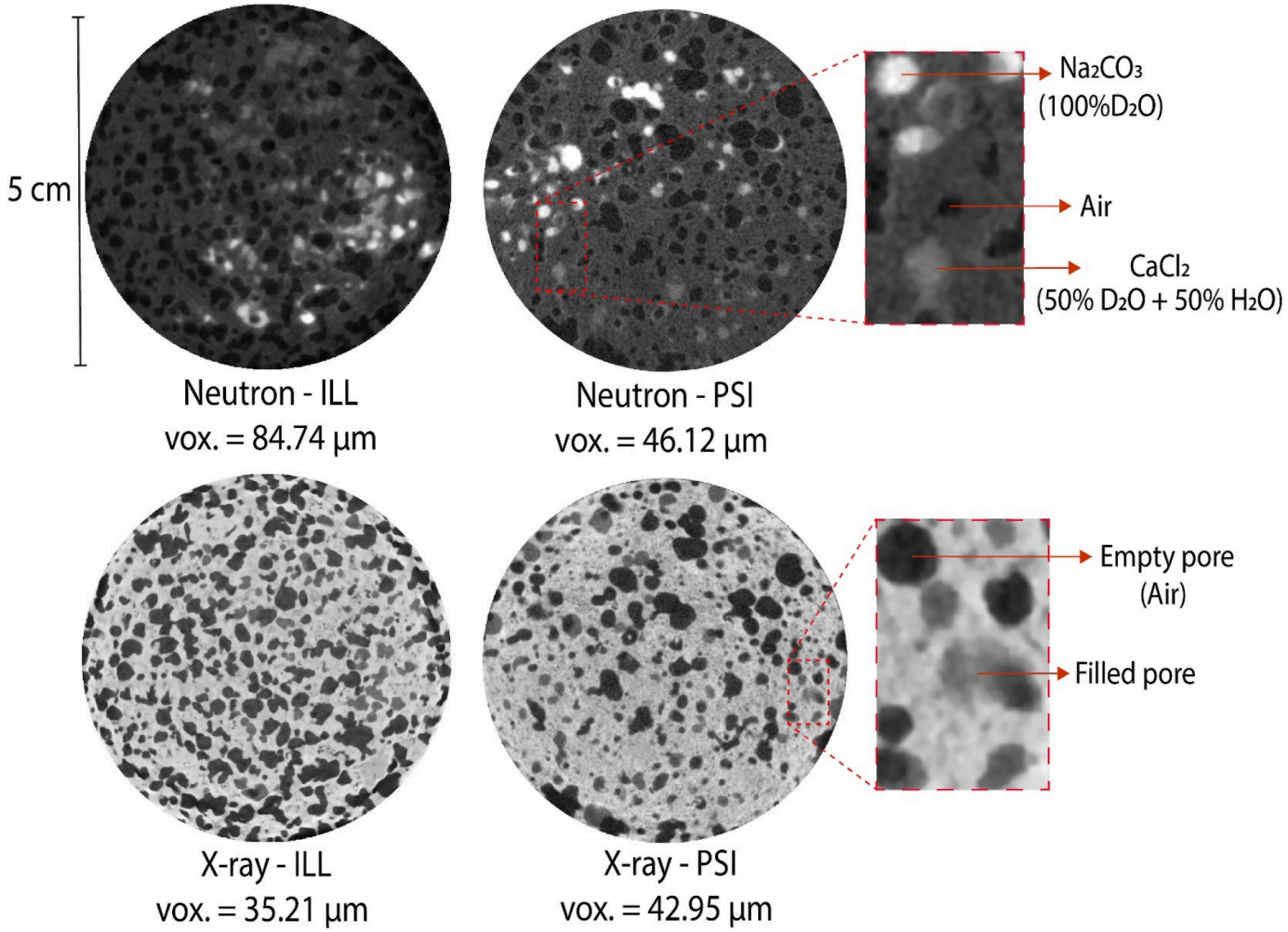


Connected, directional pathways

Same total porosity · different pore geometry

Predict the relative… **elastic compliance · anisotropy · permeability · pressure sensitivity**

Equal porosity can still produce different behavior



Neutron and X-ray tomograms showing horizontal cross-section of basalt rock core samples

Neutron images allow the separation of the content of the pores, air, and the two different solutions, whereas X-ray images provide a good contrast for the microstructure of the solid

Shafabakhsh, P., Cordonnier, B., Le Borgne, T., Mathiesen, J., Linga, G., Pluymakers, A., et al. (2025). Coupling Neutron and X-Ray Imaging of Fluid Mixing and Precipitation in Rocks: Challenges and Opportunities. *Water Resources Research*, *61*(12), e2025WR041911. https://doi.org/10.1029/2025WR041911

Small mass differences can control the answer

$$\varphi e \propto (m_sat - m_d)$$

For compact rocks, this difference can be only a few tenths of a gram.

- incomplete saturation
- surface water
- trapped bubbles
- rough geometry
- scale mismatch

Which error biases φe high? Which biases it low?

INDIVIDUAL · 3 MINUTES

Exit ticket: calculate, explain, limit

1	CALCULATE	Which mass difference gives accessible pore volume?
2	EXPLAIN	Why can equal porosity give different permeability or wave speed?
3	LIMIT	Why might a core not represent the rock mass?

Submit before leaving.

TODAY'S TAKEAWAY

A number becomes evidence only when its method, scale and uncertainty are explicit.

Next

How do forces and deformation become stress and strain?

Paterson & Wong · §2.1